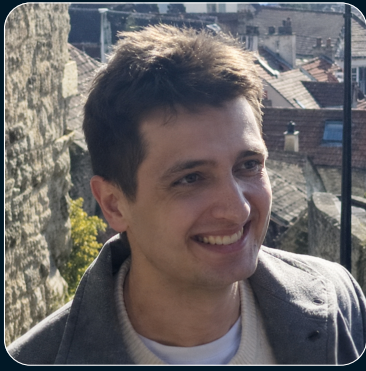




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Research Interests

Graph & Hypergraph Generation,
Generative AI, Flow-Matching,
Hierarchical Generative Models,
Reinforcement Learning for
Combinatorial Optimization,
Electronic Design Automation

Technical Skills

Languages

Python, C++, Java, C

ML & Scientific

PyTorch, PyTorch Geometric, Net-
workX, NumPy, Pandas, Matplotlib

Data

Spark, PostgreSQL

Infrastructure

Git, Docker, Linux, AWS

Academic Service

Teaching Assistant

Machine Learning with Graphs

Reviewer

NeurIPS 2025 Workshop on New
Perspectives in Graph ML

Organizer

ELLIS Doctoral Symposium
AI & Sustainability

Languages

French — Native

English — C1

Spanish — B2

Italian — B2

Interests

Arts & Culture

Cinema, Art History, Humanities

Sports

Bouldering, Running, MMA

Dorian Gailhard

PhD Student in Machine Learning

I am a third-year PhD student in Artificial Intelligence at [Télécom Paris](#) (LTCI, COM-ELEC), with a defense expected in May 2027. My research focuses on scalable machine learning for generative modeling and combinatorial optimization on graphs and hypergraphs, with applications to Electronic Design Automation (EDA).

Publications

Feature-Aware (Hyper)graph Generation via Next-Scale Prediction — ICML 2026

[Dorian Gailhard](#), [Enzo Tartaglione](#), [Lirida Naviner](#), [Jhony H. Giraldo](#) | [PDF](#) [Code](#)

Introduces the first hierarchical generative framework for graphs and hypergraphs that can generate both topology and features. By extending coarse-to-fine generation beyond topology alone, it enables the generation of attributed graphs and hypergraphs at scales beyond those supported by previous methods.

HYGENE: A Diffusion-Based Hypergraph Generation Method — AAAI 2025

[Dorian Gailhard](#), [Enzo Tartaglione](#), [Lirida Naviner](#), [Jhony H. Giraldo](#) | [PDF](#) [Code](#)

Introduces the first diffusion-based generative model for hypergraphs. By extending hierarchical graph generation to higher-order structures, it provides a scalable alternative to directly modeling the exponentially large space of possible hyperedges.

Research Experience

PhD Student — May 2024 – Present

[Télécom Paris](#) — France | [Project Page](#)

Advisors: Jhony H. Giraldo, Enzo Tartaglione, Lirida Naviner

Researching scalable generative models for graphs and hypergraphs, with applications to Electronic Design Automation. Developed hierarchical generation methods based on diffusion and flow matching, leading to first-author publications at AAAI 2025 and ICML 2026. Current work focuses on reinforcement learning for combinatorial optimization in EDA.

Research Intern — May 2023 – October 2023

[Università degli Studi di Padova](#) — Italy | [Project Page](#)

Generalized continuous formulations of variants of the Maximum Clique Problem to k -graphs. Investigated Lovász extensions and continuous relaxations for other discrete combinatorial optimization problems, proved theoretical results, and developed Python and C++ implementations of the proposed methods.

Education

M2 MVA — Mathematics, Vision, Learning — 2022–2023

[ENS Paris-Saclay](#) — France | [Coursework & Projects](#)

Research-oriented program covering machine learning, artificial intelligence, optimization, computer vision, and applied mathematics.

Master of Science — Data Science — 2021–2023

[Eurecom](#) — France | [Coursework & Projects](#)

Engineering-oriented program covering data science, machine learning, statistical learning, cloud computing, and large-scale data processing.

Engineering Degree — 2020–2023

[Télécom Paris](#) — France | [Projects](#)

Broad engineering curriculum with a specialization in computer science, covering algorithms, software engineering, mathematics, signal processing, and telecommunications.

Preparatory Classes MPSI-MP* — 2018–2020

[Lycée Hoche](#) — France

Intensive curriculum in mathematics, physics, and computer science.